

TrES-5b

R-0933

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_250713 · created 2026-07-29T11:30:53+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460869.7506 +/- 0.0042 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.124 +/- 0.047
Transit depth (Rp/Rs)^2	1.54 +/- 1.16 [%]
Orbital Inclination (inc)	80.52 +/- 2.91
Airmass coefficient 1 (a1)	0.759 +/- 0.039
Airmass coefficient 2 (a2)	0.23 +/- 0.084
Scatter in the residuals of the lightcurve fit is	5.23 %
Best Comparison Star	#2 - [486, 304]
Optimal Aperture	2.83
Optimal Annulus	11.0
Transit Duration (day)	0.033 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.124 $\pm$ 0.047 vs 0.142 (deviation 0.38 σ)
Duration fit vs published	0.033 d vs 0.0758 d (deviation 1.48 σ)
Mid-time O-C	-10.7 min
Depth SNR	2.6
Residual scatter	5.23 %

Light curve

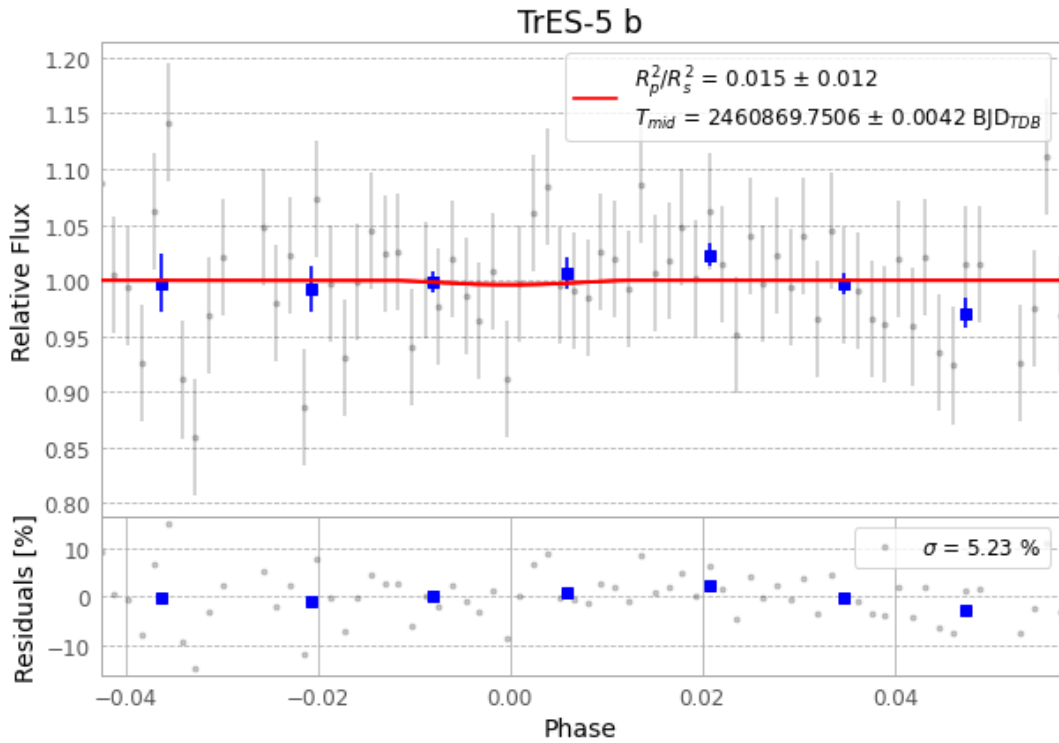


Figure 1 — FinalLightCurve_TrES-5 b_2025-07-12.png (SHA-256 2ad5c4f2b645f0c0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [351, 223], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 b9f016301724a97d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

72 FITS frames (47 MB), TRES-5250713042715.FITS → TRES-5250713080016.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-250713.log (6c59a7755cd8...), FinalLightCurve_TrES-5 b_2025-07-12.png (2ad5c4f2b645...), FinalLightCurve_TrES-5 b_2025-07-12.csv (c0735d01946a...)

Compute resources

Wall time 115.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 11:30Z.